

Circular Statistics - Quick Reference

SEAC 2026 Workshop: Circular Statistics in Archaeoastronomy and Ethnoastronomy Using R

1. Why circular statistics?

Directions repeat after one complete turn. On a 0-360 degree scale, 359 degrees and 1 degree are only 2 degrees apart, even though ordinary arithmetic treats them as far apart.

Teaching example:

358, 359, 1, 2 degrees

The ordinary arithmetic mean is 180 degrees, which points South. The circular mean is approximately 0/360 degrees, which correctly represents the cluster around North.

2. Directional or axial?

Directional data: start and end matter. Example: orientation measured from a church entrance toward the apse.

Axial data: the axis has no arrow. An axis at 20 degrees is equivalent to the same axis at 200 degrees.

Never analyse axial measurements as ordinary 0-360 degree directional data without an appropriate transformation.

3. Geographic azimuth convention

For this workshop:

- 0 degrees = North
- 90 degrees = East
- 180 degrees = South
- 270 degrees = West
- angles increase clockwise.

Always document the convention used.

4. Circular object in R

```
library(circular)

az <- circular(
  dat$azimuth_deg,
  units = "degrees",
  template = "geographics",
  modulo = "2pi"
)
```

5. Rose diagram

```
rose.diag(
  az,
  bins = 18,
  radii.scale = "sqrt"
)
```

Plot first, test second. Try more than one defensible bin count.

6. Mean direction

```
mean_deg <- as.numeric(mean(az)) %% 360
```

Use only when a single central direction is meaningful. A pooled or multimodal sample can make one mean misleading.

7. Mean resultant length (R-bar)

```
Rbar <- rho.circular(az)
```

- near 1: strong net directional concentration;
- near 0: weak net directional concentration.

Important: R-bar near 0 can result from uniformity **or from opposing clusters that cancel vectorially**.

8. Circular variance

```
circular_variance <- 1 - Rbar
```

- near 0: tighter concentration;
- larger values: more dispersion.

9. Circular standard deviation

With the current workshop workflow:

```
circular_sd_deg <- sd(az) * 180 / pi
```

The explicit conversion keeps the reported result in degrees.

10. Confidence interval for mean direction

```
set.seed(2026)
```

```
mean_ci <- mle.vonmises.bootstrap.ci(  
  az,  
  alpha = 0.05,  
  reps = 2000  
)
```

Use the interval as a measure of precision for the estimated mean direction under the bootstrap/model assumptions. It does not measure archaeological certainty.

11. Rayleigh test

```
rayleigh_result <- rayleigh.test(az)
```

```
Rbar <- rho.circular(az)  
z <- length(az) * Rbar^2  
p <- rayleigh_result$p.value
```

H0: directions are uniform around the circle.

General alternative: unimodal directional concentration.

A small p-value is evidence against circular uniformity for that alternative. A non-significant result does not prove uniformity, especially for bimodal or multimodal patterns.

12. von Mises model

The von Mises distribution is a useful single-cluster circular model.

- μ = central direction
- κ = concentration; larger values indicate tighter concentration.

```
vm <- mle.vonmises(az)
```

```
as.numeric(vm$mu) %% 360
```

```
vm$kappa
```

Do not force a single von Mises model onto an obviously multimodal sample.

13. Curvigram baseline

For the workshop declination exercise:

```
Coordinate:      declination_deg
Geometry:        linear
Kernel:          Epanechnikov
Bandwidth:       2 x uncertainty_deg
Scaling:         area = 1
Range:           -45 to +45 degrees
Step:            0.05 degrees
```

`uncertainty_deg` is the reported plus/minus measurement uncertainty. It is not automatically assumed to be one standard deviation.

14. Curvigram sensitivity checks

At minimum compare:

```
Epanechnikov x 1
Epanechnikov x 2    baseline
Epanechnikov x 3
Gaussian x 2
```

Ask whether the substantive feature stays in approximately the same place and remains visible.

15. Three rules to remember

1. **Plot first, test second.**
2. **Match the method to the distribution and research question.**
3. **Statistical concentration is not proof of cultural or astronomical intention.**